packages/graph-core/src/fingerprint.ts

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Overview

`packages/graph-core/src/fingerprint.ts` replaces the previous ad-hoc change-detection logic with a structured fingerprinting system for source files.

Key changes

- **Imports added** (lines 1-5): `node:crypto`, `node:fs`, `node:path`, `./types.js`, `./plugins/registry.js`.
- **New types** (lines 9-63): `FunctionFingerprint`, `ClassFingerprint`, `ImportFingerprint`, `FileFingerprint`, `FingerprintStore`, `ChangeLevel`, `FileChangeResult`, `ChangeAnalysis`.
- **`contentHash`** (lines 70-72) computes a SHA-256 hash of file contents.
- **`extractFileFingerprint`** (lines 79-122) builds a structural fingerprint from a `StructuralAnalysis`, capturing functions, classes, imports, exports, and line counts.
- **`compareFingerprints`** (lines 131-246) compares two fingerprints, classifying changes as `NONE`, `COSMETIC`, or `STRUCTURAL` and producing a diff list.
- **`buildFingerprintStore`** (lines 253-291) creates a fingerprint store for a project, falling back to content-hash-only fingerprints when structural analysis is unavailable (lines 268-281).
- **`analyzeChanges`** (lines 297-385) orchestrates change detection across a set of files, returning a `ChangeAnalysis` summary.

Impact

- **Correctness**: Deterministic structural change detection; conservative `STRUCTURAL` classification when analysis is missing (lines 142-149).
- **Maintainability**: Centralized fingerprint logic and explicit type definitions aid future extensions.
- **Performance**: Adds file I/O and hashing per file; early hash comparison (lines 138-140) mitigates unnecessary work, but overall runtime depends on project size.

Risks & follow-ups

- **False positives**: Structural comparison may flag benign changes (e.g., reordered imports) as `STRUCTURAL` (lines 220-226).
- **Missing analysis**: Files without tree-sitter support are always marked `STRUCTURAL` (lines 268-281); confirm this aligns with business expectations.
- **Registry integration**: `registry.analyzeFile` must return a `StructuralAnalysis`; test that unsupported file types are handled correctly.
- **Performance regression**: Benchmark `buildFingerprintStore` on large codebases to ensure acceptable runtime.